Class 9 Mathematics — Exercise 5.4

Basic Concept

(1) Inequality (Inequation)

A statement that **"two algebraic expressions are not equal"** is called an inequality or inequation.

Example: $5y - 7 < 3y + 9$

(2) Inequality Symbols

Symbol	Meaning	Example
$<$	is less than	$-5 < 7$
$>$	is greater than	$10 > -3$
$\leq$	is less than or equal to	$7 \leq 7$
$\geq$	is greater than or equal to	$5 \geq 1$
$\neq$	is not equal to	$3 \neq 5$

(3) Linear Inequality in One Variable

An inequality having only one variable with power 1, e.g.

$$\text{(i)}\ \ 5y - 7 < 3y + 9 \qquad\qquad \text{(ii)}\ \ x \leq 3$$

(4) Important Rule

When both sides of an inequality are **multiplied or divided by a negative number**, the **direction of the inequality sign is reversed**.

$$5 < 10 \quad \xrightarrow{\times(-1)} \quad -5 > -10$$

(5) Compound Inequalities

Two inequalities joined by the word **"and"** or **"or"**, e.g. $2x < 6$ and $3x + 2 > -4$; $3x + 5 > 7$ or $4x - 1 < 3$

- **"or":** solve both separately; the solution set is the **union** of both solution sets.
- **"and":** solve both separately; the solution set is the **intersection** of both solution sets.

(6) Useful Sets

$$N = \{1, 2, 3, 4, 5, ...\} \qquad W = \{0, 1, 2, 3, 4, ...\} \qquad R = \text{real numbers}$$

Question 1(a)

Check whether the given value of each variable satisfies the inequality.

(i) $5y - 12 > 0; \quad y = 3$

$$5(3) - 12 > 0$$
$$15 - 12 > 0$$
$$3 > 0$$

Yes, it satisfies the inequality.

(ii) $4 - 2x \leq 9; \quad x = -3$

$$4 - 2(-3) \leq 9$$
$$4 + 6 \leq 9$$
$$10 \leq 9$$

No, it does not satisfy the inequality.

(iii) $5 - 2x > -4x + 5; \quad x = 4$

$$5 - 2(4) > -4(4) + 5$$
$$5 - 8 > -16 + 5$$
$$-3 > -11$$

On the number line, -3 lies to the right of -11, so $-3 > -11$ is true.

Yes, it satisfies the inequality.

(iv) $3(z + 4) \leq 6; \quad z = -2$

$$3(-2 + 4) \leq 6$$
$$3(2) \leq 6$$
$$6 \leq 6$$

Yes, it satisfies the inequality.

(v) $5(x - 2) \geq 9x - 3(2x - 4); \quad x = 11$

$$5(11 - 2) \geq 9(11) - 3[2(11) - 4]$$
$$5(9) \geq 99 - 3(22 - 4)$$
$$45 \geq 99 - 3(18)$$
$$45 \geq 99 - 54$$
$$45 \geq 45$$

Yes, it satisfies the inequality.

Question 1(b)

Write each solution in the set notation form.

(i) $2 \leq x \leq 5$, **where** $x \in N$

$N = \{1, 2, 3, 4, 5, ...\}$

$$\boxed{\{2, 3, 4, 5\}}$$

(ii) $y < 7$, **where** $y \in N$

$N = \{1, 2, 3, 4, 5, ...\}$

$$\boxed{\{1, 2, 3, 4, 5, 6\}}$$

(iii) $z \leq -3$, **where** $z \in R$

$$\boxed{\{z \mid z \in R \wedge z \leq -3\}}$$

(iv) $x \leq 4$, **where** $x \in W$

$W = \{0, 1, 2, 3, 4, ...\}$

$$\boxed{\{0, 1, 2, 3, 4\}}$$

(v) $-4 < x < -\frac{3}{2}$, **where** $x \in R$

$$\boxed{\left\{x \mid x \in R \wedge -4 < x < -\frac{3}{2}\right\}}$$

Question 2

$$3x - 2 < 7, \quad x \in N$$

Solution:

$$3x < 7 + 2$$
$$3x < 9$$
$$x < \frac{9}{3}$$
$$\boxed{x < 3}$$

As $x \in N$, $\quad N = \{1, 2, 3, 4, 5, ...\}$

$$\boxed{S.S = \{1, 2\}}$$

Question 3

$$6x - 5 \leq 35 - 2x, \quad x \in W$$

Solution:

$$6x + 2x \leq 35 + 5$$
$$8x \leq 40$$
$$x \leq \frac{40}{8}$$

$$\boxed{x \leq 5}$$

As $x \in W, \quad W = \{0, 1, 2, 3, 4, ...\}$

$$\boxed{S.S = \{0, 1, 2, 3, 4, 5\}}$$

Question 4

$$16 - 5y < 4(y - 1) - 7, \quad y \in R$$

Solution:

$$16 - 5y < 4y - 4 - 7$$
$$-5y - 4y < -4 - 7 - 16$$
$$-9y < -27$$

Dividing by -9 (negative), so the sign is reversed:

$$y > \frac{-27}{-9}$$
$$\boxed{y > 3}$$

As $y \in R$:

$$\boxed{S.S = \{y \mid y \in R \wedge y > 3\}}$$

Question 5

$$10 - (7 - y) \geq 3y - 9, \quad y \in R$$

Solution:

$$10 - 7 + y \geq 3y - 9$$
$$y - 3y \geq -9 - 10 + 7$$
$$-2y \geq -12$$

Dividing by -2 (negative), so the sign is reversed:

$$y \leq \frac{-12}{-2}$$
$$\boxed{y \leq 6}$$

As $y \in R$:

$$\boxed{S.S = \{y \mid y \in R \wedge y \leq 6\}}$$

Question 6

$$5 - 3x < 11 \quad \text{or} \quad 2x + 3 < -9$$

Solution:

$5 - 3x < 11$	$2x + 3 < -9$
$-3x < 11 - 5$	$2x < -9 - 3$
$-3x < 6$	$2x < -12$
$x > \frac{6}{-3}$	$x < \frac{-12}{2}$
$\boxed{x > -2}$	$\boxed{x < -6}$

As $x \in R$:

$$\boxed{S.S = \{x \mid x \in R \wedge x > -2 \text{ or } x < -6\}}$$

OR

$$\{x \mid x \in R \wedge -6 > x \text{ or } x > -2\}$$

Question 7

$$2x + 3 \leq 9 \quad \text{and} \quad x - 5 > -6$$

Solution:

$2x + 3 \leq 9$	$x - 5 > -6$
$2x \leq 9 - 3$	$x > -6 + 5$
$2x \leq 6$	$\boxed{x > -1}$
$x \leq \frac{6}{2}$	
$\boxed{x \leq 3}$	

As $x \in R$:

$$\boxed{S.S = \{x \mid x \in R \wedge -1 < x \leq 3\}}$$

Question 8

$$1 \leq 7 - 3x \leq 22$$

Solution: Split into two inequalities (both must hold):

$$1 \leq 7 - 3x \quad \text{and} \quad 7 - 3x \leq 22$$

$1 \leq 7 - 3x$	$7 - 3x \leq 22$
$3x \leq 7 - 1$	$-3x \leq 22 - 7$
$3x \leq 6$	$-3x \leq 15$
$x \leq \frac{6}{3}$	$x \geq \frac{15}{-3}$
$\boxed{x \leq 2}$	$\boxed{x \geq -5}$

As $x \in R$:

$$\boxed{S.S = \{x \mid x \in R \wedge -5 \leq x \leq 2\}}$$

Question 9

$$3x + 21 < 1 - x \quad \text{or} \quad 3x + 8 > 3 - 2x$$

Solution:

$3x + 21 < 1 - x$	$3x + 8 > 3 - 2x$
$3x + x < 1 - 21$	$3x + 2x > 3 - 8$
$4x < -20$	$5x > -5$
$x < \frac{-20}{4}$	$x > \frac{-5}{5}$
$\boxed{x < -5}$	$\boxed{x > -1}$

As $x \in R$:

$$\boxed{S.S = \{x \mid x \in R \wedge x < -5 \text{ or } x > -1\}}$$

Question 10

$$1 - 5x > 16 \quad \text{and} \quad 3 - \frac{3x}{2} \leq 9$$

Solution:

$1 - 5x > 16$	$3 - \frac{3x}{2} \leq 9$
$-5x > 16 - 1$	$-\frac{3x}{2} \leq 9 - 3$
$-5x > 15$	$-\frac{3x}{2} \leq 6$
$x < \frac{15}{-5}$	$-3x \leq 12$
$\boxed{x < -3}$	$x \geq \frac{12}{-3}$

$1 - 5x > 16$	$3 - \frac{3x}{2} \leq 9$
	$\boxed{x \geq -4}$

As $x \in R$:

$$\boxed{S.S = \{x \mid x \in R \wedge -4 \leq x < -3\}}$$

Question 11

The sum of five times a number x and 10 is less than -35 or greater than -5. What real numbers does x represent?

Solution:

$$5x + 10 < -35 \quad \text{or} \quad 5x + 10 > -5$$

$5x + 10 < -35$	$5x + 10 > -5$
$5x < -35 - 10$	$5x > -5 - 10$
$5x < -45$	$5x > -15$
$x < \frac{-45}{5}$	$x > \frac{-15}{5}$
$\boxed{x < -9}$	$\boxed{x > -3}$

$$\boxed{S.S = \{x \mid x \in R \wedge x < -9 \text{ or } x > -3\}}$$

Question 12

Two times a number decreased by 5 is greater than or equal to the number increased by 8. Find the possible values for the number.

Solution: Let the number be x.

$$2x - 5 \geq x + 8$$
$$2x - x \geq 8 + 5$$
$$\boxed{x \geq 13}$$

$$\boxed{S.S = \{x \mid x \in R \wedge x \geq 13\}}$$